IBM Db2 TechX Summit 2026 — Db2Hack 24-Hour Challenge

FinSight AI

Database-First Personal Financial Intelligence & Anomaly Reconciliation Platform

Tagline: "Your money has a story. Understand it."

Hackathon Event	IBM Db2Hack 2026 (GL Bajaj, Mathura in association with IBM & NexaSphere)
Submission Track	AI Applications on IBM Db2 (Relational + Native Vector Search)
Core Stack	IBM Db2 Community Edition 12.1.5.0, LangChain, FastEmbed ONNX, Rust Axum, React 19
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Executive Summary

FinSight AI is a database-first personal financial intelligence platform engineered to eliminate generative AI hallucinations in personal finance. Instead of streaming sensitive, unredacted financial records to third-party language models, FinSight AI anchors all financial truth, semantic indexing, and mathematical aggregations directly inside IBM Db2 12.1.5.0. Utilizing native Db2 VECTOR(384, FLOAT32) storage and VECTOR_DISTANCE cosine scoring alongside LangChain tool orchestration, the system delivers grounded financial queries, explainable anomaly detection, conceptual transaction search, and secure statement ingestion. A memory-safe Rust Axum API gateway guarantees strict multi-tenant isolation and cryptographic CSRF protection.

Project Vision & Guiding Invariant

In financial software, **arithmetic accuracy and auditability are non-negotiable**. Language models are outstanding synthesis engines but notorious arithmetic hallucinators. FinSight AI strictly separates reasoning from mathematics: IBM Db2 performs 100% of the ledger calculations, aggregations, and vector distance rankings. The AI model only receives verified database results, ensuring zero hallucinated balances.

1. Real-World Problem Statement

Modern consumer and small business finance is characterized by fragmented transactions distributed across credit cards, UPI payments, net banking portals, and digital subscription services. Despite widespread digital payments, users lack actionable visibility into their true financial position.

The Three Critical Deficiencies of Existing Systems:

- **Opaque & Uninformative Statement Ledgers:** Traditional online banking portals display chronological tables of cryptic merchant codes (e.g. ACH-94821 AWS-BILLING-SEATTLE) without contextual understanding, recurring bill identification, or multi-month cash flow trajectories.
- **The LLM Hallucination Trap in FinTech:** Generic AI wrappers pass transaction dumps to an LLM context window. Language models routinely fabricate transaction amounts, miscalculate monthly net surpluses, and invent dates. In personal finance, an incorrect balance figure is catastrophic.
- **Data Privacy, Sovereignty & Multi-Tenant Insecurity:** Streaming unredacted banking records across external cloud LLM APIs violates financial confidentiality and exposes users to prompt injection or cross-tenant data leaks.

2. Proposed Solution & Target Audience

FinSight AI pioneers a **Database-First Architectural Paradigm**. Rather than asking an AI to compute financial totals, IBM Db2 12.1.5.0 acts as the authoritative calculation engine and vector store. The AI agent acts strictly as an explainable orchestrator that translates user intent into audited SQL and vector queries.

Target User Personas:

- **Salaried Professionals:** Individuals managing monthly living expenses, recurring subscriptions, and savings targets who need instant, accurate answers to questions like *'What did I spend on dining out in October?'* without manual spreadsheet entry.
- **Freelancers & Contractors:** Users with irregular income streams and complex business deductions who require automated anomaly alerts and transaction deduplication.
- **Small Business Operators:** Owners needing printable, audit-ready monthly financial statements and category trajectory analyses without the overhead of enterprise accounting suites.

3. Key Product Capabilities (Implemented & Verified)

FinSight AI provides six core, production-tested capabilities anchored directly in IBM Db2 12.1.5.0:

1. Executive Financial Command Center

The authenticated dashboard acts as the user's financial command post. It delivers: (1) an authoritative Net Operating Surplus hero card computed from verified Db2 credits minus debits; (2) a 6-month historical cash flow trajectory visualization contrasting inflows and outflows; (3) dynamic category allocation charts; and (4) rapid pagination over the most recent ledger entries. All calculations strictly utilize Db2 DECIMAL(14,2) precision.

2. Dual-Mode Grounded Financial Assistant

The conversational assistant implements a resilient, dual-engine design that solves the hallucination problem:

• **Mode A (Deterministic Db2 SQL Engine):** Executes pre-compiled, audited SQL queries directly in Db2. Delivers 0% hallucination risk, sub-50ms latency, and transparent SQL proof citations.

• **Mode B (LangChain + Gemini Engine):** Uses LangChain dynamic tool calling to bind 10 deterministic Db2 tools to Google Gemini 3.1 Flash-Lite, generating natural language summaries with verified transaction ID citations.

• **Defensive Fallback:** If Mode B encounters API rate limiting (such as the current Google Free Tier HTTP 429 quota exhaustion), the service automatically routes the request to Mode A without server exceptions or user interruption.

3. Native IBM Db2 Semantic Transaction Search

Traditional search requires exact substring matches. FinSight AI enables conceptual natural language search: users can search for *'daily office commute'*, *'cloud hosting infrastructure'*, or *'takeout and coffee'*. Queries are converted into 384-dimensional dense vectors via a local FastEmbed ONNX runtime and evaluated in-database using native Db2 VECTOR_DISTANCE(..., 'COSINE') queries. Nearest neighbor transactions are retrieved with exact distance scores.

3. Key Product Capabilities (Continued)

4. Explainable Anomaly Intelligence

FinSight AI rejects opaque 'black box' risk scores in favor of transparent, rule-and-heuristic anomaly detection grounded in Db2:
• **Recurring Subscription Drift:** Detects severe surges in regular software or service bills. In the demo dataset, AWS billing jumped from ■1,380/month to ■42,800 (+3,014% surge), flagged with a 91% confidence score and statistical rationale.
• **Velocity Outliers:** Identifies bursts of high-frequency debits within a narrow settlement window.
• **Duplicate Authorizations:** Highlights charges with identical amounts, dates, and merchants.
• **Actionable Workflow:** Users can mark alerts as REVIEWED or DISMISSED, updating the FINSIGHT.ANOMALIES table in Db2.

5. Secure CSV Ingestion & Deduplication Wizard

Users can upload historical bank statements through an interactive 4-step wizard:
1. **File Validation:** Client- and gateway-enforced 2MB file limits, MIME type verification, and header checking.
2. **Intelligent Column Mapping:** Maps diverse CSV layouts (e.g. separate debit/credit columns vs. signed amount).
3. **Duplicate Pre-Check:** Compares parsed records against existing transactions in Db2 to prevent double-counting.
4. **Committed to Db2 & Vectorized:** Inserts valid records into FINSIGHT.TRANSACTIONS and automatically indexes 384-dimensional dense vectors.

6. Financial Insights & Monthly Audit Reports

Provides month-over-month expenditure shift metrics, net savings ratios, and category trajectories:
• **MoM Comparison:** Demonstrates that November living expenditures dropped by 72.8% compared to October (when a large electronics purchase occurred).
• **Printable Financial Statement:** Formats a professional executive audit report with income breakdown, top 5 merchants, and savings metrics ready to print or archive as PDF.

4. Agentic AI Framework: LangChain Integration

FinSight AI leverages LangChain (langchain-core, langchain-google-genai, langchain-community) as an explicit agentic tool-orchestration harness. Rather than relying on autonomous, unconstrained multi-agent swarms that risk unpredictable loops, FinSight AI employs a strictly bounded, deterministic tool-calling pattern.

Ten Structured IBM Db2 Financial Tools:

The Python AI service registers 10 audited tools bound directly to Db2 SQL queries:
1. DB2_TOOL_GET_INCOME — Extracts verified credit deposits for a specified month/year.
2. DB2_TOOL_LARGEST_EXPENSES — Retrieves top debit transactions ranked by amount.
3. DB2_TOOL_MONTHLY_CASH_FLOW — Computes multi-month inflows, outflows, and net delta.
4. DB2_TOOL_SUBSCRIPTIONS — Aggregates recurring charges and highlights anomalous surges.
5. DB2_TOOL_ANOMALY_ALERTS — Retrieves active statistical outliers from FINSIGHT.ANOMALIES.
6. DB2_TOOL_PENDING_TRANSACTIONS — Isolates unposted authorizations awaiting settlement.
7. DB2_TOOL_CATEGORY_BREAKDOWN — Categorical spending distribution and percentage of total.
8. DB2_TOOL_TOP_MERCHANTS — Aggregates expenditure volume by merchant name.
9. DB2_TOOL_SEARCH_TRANSACTIONS — Server-side text matching with pagination.
10. DB2_TOOL_VECTOR_SIMILARITY — In-database semantic distance ranking.

LangChain Vector Store Wrapper (DB2VS):

FinSight AI implements a custom LangChain VectorStore interface (DB2VS) bridging LangChain's vector search semantics directly into native IBM Db2 12.1.5.0 SQL. This decouples the embedding generator from the database, allowing any LangChain retriever or chain to seamlessly query Db2 vector columns.

Dual-Engine Fallback Resilience:

A critical engineering requirement in fintech is high availability. When external cloud LLMs encounter HTTP 429 quota exhaustion or network latency, FinSight AI's LangChain assistant wrapper catches the exception and immediately invokes Mode A (Deterministic SQL). This ensures that user queries always receive 100% accurate, SQL-grounded answers without downtime.

5. IBM Db2 12.1.5.0 Deep Integration & Vector Search

IBM Db2 Community Edition (Linux/x86_64 v12.1.5.0, database FINDB) acts as the unified transactional and vector engine. By consolidating relational accounting records and dense vector embeddings into a single database instance, FinSight AI eliminates the latency, operational complexity, and data drift inherent in maintaining separate relational and vector databases.

Native Db2 Vector Implementation:

• **Native Vector Type:** The FINSIGHT.TRANSACTIONS table features a native VECTOR(384, FLOAT32) column. Embeddings are computed locally using FastEmbed's BAAI/bge-small-en-v1.5 model in ONNX Runtime, eliminating external API costs.
• **In-Database Cosine Calculation:** Semantic searches execute entirely in Db2 SQL using the native VECTOR_DISTANCE function:
SELECT transaction_id, transaction_date, merchant_name, amount, category, VECTOR_DISTANCE(embedding, CAST(? AS VECTOR(384, FLOAT32)), 'COSINE') AS distance FROM FINSIGHT.TRANSACTIONS WHERE user_id = ? ORDER BY distance ASC FETCH FIRST 10 ROWS ONLY

Db2 Schema Architecture:

• FINSIGHT.USERS — User accounts, email constraints, and Argon2id password hashes.
• FINSIGHT.SESSIONS — Server-side session tokens, user references, and TTL expiry timestamps.
• FINSIGHT.TRANSACTIONS — Authoritative ledger with DECIMAL(14,2) precision, direction, category, and native vector column.
• FINSIGHT.ANOMALIES — Scored anomaly alerts with severity, confidence score, and review workflow status.

Audited Ledger Reconciliation in Db2:

A baseline audit of the 208 seeded transactions demonstrates perfect mathematical reconciliation in Db2:
• **Total Posted Credits (Salary Inflows, 6 months):** 6 transactions | ■750,000.00
• **Total Posted Debits (Living Expenditures):** 190 transactions | ■500,162.00
• **Total Posted Debits (Internal Account Transfers):** 6 transactions | ■210,000.00
• **Total Pending Debits (Unsettled Authorizations):** 6 transactions | ■6,717.00
• **Reconciled Net Operating Surplus: ■249,838.00** (■750k - ■500,162).

6. System Architecture & End-to-End Workflow

FinSight AI employs a decoupled, four-tier architecture designed for defense-in-depth, zero secret leakage, and memory safety:

```
+-------------------------------------------------------------------------------------+ | PRESENTATION LAYER (CLIENT
SPA) | | React 19 + TypeScript + Vite + Tailwind CSS (Forest Green & Butter Cream) |
+-------------------------------------------------------------------------------------+ | HTTPS / Cookie + HMAC CSRF
v +-------------------------------------------------------------------------------------+ | GATEWAY LAYER (RUST AXUM
:8080) | | Argon2id Auth · Constant-Time CSRF Verification · Token-Bucket Rate Limiting (429) |
+-------------------------------------------------------------------------------------+ | Internal HTTP / Bearer
Auth v +-------------------------------------------------------------------------------------+ | AI & INGESTION
SERVICE (FASTAPI :8000) | | FastEmbed ONNX (384-Dim) · LangChain 10-Tool Engine · Dual-Mode Fallback Router |
+-------------------------------------------------------------------------------------+ | ibm_db TCP / Parameterized
SQL v +-------------------------------------------------------------------------------------+ | DATABASE LAYER (IBM
DB2 12.1.5.0) | | FINSIGHT.TRANSACTIONS (DECIMAL 14,2 + VECTOR 384) · VECTOR_DISTANCE(COSINE) · SESSIONS|
+-------------------------------------------------------------------------------------+
```

End-to-End Workflow Traces:

1. **Client Authentication:** User submits credentials to Rust gateway. The gateway verifies Argon2id hash against Db2, generates a 64-char session token stored in FINSIGHT.SESSIONS, and sets an HttpOnly, SameSite=Lax cookie with deterministic HMAC-SHA-256 CSRF protection.

2. **AI Assistant Query:** User asks: *'What was my total income in October?'*. The Rust gateway validates session and CSRF headers, injects verified X-User-Id, and proxies to Python AI service. In Mode B, LangChain invokes DB2_TOOL_GET_INCOME. In Mode A or during Gemini quota limits, Db2 SQL executes directly, returning ■125,000.00 with transparent citations.

3. **Semantic Search:** User searches *'food delivery and cafes'*. FastAPI encodes query to 384-dim vector using FastEmbed ONNX, executes VECTOR_DISTANCE in Db2, and returns Swiggy/Zomato records ranked by cosine proximity.

7. Impact, Technology Stack & Prototype Disclosures

Technology Stack Breakdown:

Layer	Technologies	Architectural Role
Database	IBM Db2 v12.1.5.0 (FINDB)	Financial ledger, DECIMAL(14,2), native VECTOR(384) & cosine search.
Gateway	Rust 1.85, Axum 0.7, Argon2id	Memory-safe ingress, session validation, HMAC CSRF, rate limiting.
AI Service	Python 3.11, FastAPI, Uvicorn	LangChain 10-tool orchestration, FastEmbed ONNX, anomaly scoring.
Frontend	React 19, TypeScript, Vite, Tailwind	Authenticated command center, dual-mode assistant UI, CSV wizard.

Mandatory Hackathon Disclosures & Prototype Boundaries

1. Prototype Scope: FinSight AI is an engineering prototype submitted to IBM Db2Hack 2026; it is not commercial financial advisory software.
2. Synthetic Ledger: All 208 demo records across 6 months are synthetic data created for demo user Alex Chen. Zero real banking data is stored.
3. Statement Ingestion: Statements are imported via a verified CSV wizard with column mapping and deduplication. No live bank scraping or Open Banking APIs.
4. Gemini LLM Quota Status: The development Google AI Studio Free Tier key has exhausted its daily quota (HTTP 429). The system operates seamlessly in Mode A (Deterministic Db2 SQL Engine) without server errors or downtime.

Conclusion & Submission Attestation